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Machine Learning Approaches for EMG Signal Classification to Predict Hand Gesture

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***Abstract*— In this study, the hand gestures prediction for electromyography (EMG) signals classification is provided. EMG signals which are electrical stimulations of muscles were measured, processed and analyzed to achieve discriminative features in time, frequency and time-frequency domain. To improve a better result of the classification, a number of classifiers namely, Support Vector Machine (SVM), k-Nearest Neighbors (k-NN), Random Forest (RF) and Artificial Neural Network (ANN) have been compared with each other with the objective of hybrid feature combination to enhance the accuracy of the result of the classification. As it turned out, ANN proved to be the most precise in classification (96.2 percent), followed by SVM (94.6 percent), RF (92.7 percent) and k-NN (89.4 percent). These findings suggest that hybrid feature extraction works better in terms of discriminative power and nonlinear learning capacity of ANN can be employed to extract the complicated and non-stationary nature of EMG signals. Overall, this research paper indicates the potential of hybrid feature extraction in conjunction with sophisticated learning architectures to enhance performance, robustness and flexibility of EMG-based hand gesture recognition systems to be applied in the field of prosthetic control, rehabilitation robotics as well as human-machine interfaces.**

Keywords*— *Electromyography (EMG), Machine Learning, Hand Gesture Recognition, Artificial Neural Network (ANN), Feature Extraction.

I. INTRODUCTION

The EMG signals are essential physiological indications of human movement and neuromuscular control created due to the electrical activity of skeletal muscles [1]. EMG-based hand gesture recognition has become a revolutionary field of research in the biomedical engineering community, HCI and rehabilitation robotics in recent years [2]. EMG signal analysis provides the ability to control prosthetic limbs, assistive devices, exoskeletons and virtual interfaces by decoding muscle activation patterns into meaningful gestures and thus bridging the gap between the human intention and machine action [3]. Nonetheless, the classification of the EMG signals is associated with major challenges as it is a complex process by its nature since EMG signals are non-stationary, person-specific, and readily influenced by noise, electrode variations, and muscle fatigue [4]. Conventional rule-based or thresholding systems have been found insufficient to meet the accuracy and resilience that are needed in real-time systems, prompting the increasing use of machine learning (ML) systems as an innovation in this field. ML algorithms have the capability to learn discriminative patterns on EMG streams automatically, and classify hand gestures without using manual feature engineering. SVM, k-NN, and Random Forests as well as ANN methods of supervised learning have been demonstrated to recognize both non-kinesthetic and dynamic gestures and further deep learning networks include CNN and Recurrent Neural Networks (RNN) which are more effective in uncovering more hierarchical features in raw signals. Although these developments have been made, high accuracy on subjects and on different conditions is a vital issue because of the anatomy of the muscle, electrode set up, and signal non-linearities. The existing literature, in turn, focuses on hybrid style that combines signal processing, feature optimization and adaptive learning to enhance real-time reliability and scalability. This paper examines different machine learning modalities in identifying hand gestures through the EMG signal, as a way of lowering the error in classifying hand gestures, to achieve a useful pipeline in classifying hand gestures and aid in the advancement of rehabilitation technologies in the future.

A. Background of the study

The electromyography (EMG) is a technique of recording and analyzing electrical response produced by the skeletal muscle when they contract. The motor neurons generate electrical potentials during contraction of a muscle and this potential can be measured using either surface electrodes or intramuscular electrodes. Such EMG signals will provide good suggestions of the physiological of the muscles like the strength, fatigue level and coordination. [5]. EMG has evolved to be an important instrument in biomedical research, clinical diagnosis, and human-machine interface development due to its ability to represent the direct neural control of movement over the years. Hand gesture recognition is one of the most promising areas of the use of the EMG signals, which is the process of recognizing the patterns of muscle activity to identify the particular movements of the hands or fingers. Given that hand gestures are strongly associated with the muscle activity in the

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forearm and the wrist, the EMG signals can be utilized successfully to differentiate between various gestures, gripping, pointing, waving, or pinching. This has provided a new dimension of possibilities in a number of areas specifically in assistive and interactive technologies.

In the field of prosthetics, EMG-based gesture recognition can be used to create sophisticated myoelectric prosthetic limbs capable of decoding the muscle activity of its user to create natural gestures and reinstate autonomy and livelihood quality of amputees. Likewise, in human-computer interaction (HCI), the EMG-controlled systems enable one to operate digital devices, robot arms, or simulators by muscle-intuitive muscle-based commands without requiring any physical contact or remote controllers [6]. All these applications find application in rehab robotics, games, virtual reality, and assistive technology to assist people with motor impairments. Therefore, EMG-based hand gesture recognition is a major milestone towards the development of smooth, smart interfaces through which human will is linked to machine behavior. Through the strength of signal processing and machine learning, scientists are striving to improve the precision, flexibility, and reactivity of these systems to be implemented in reality.

B. Problem Statement

The main problem in recognition of hand gestures based on EMG is the correct classification of gestures on variable and complicated EMG signals. EMG signals are non-stationary by nature, i.e. their statistical properties are varying with time because of fatigue of their muscle, displacement of electrodes and change in skin conductivity. Moreover, such signals are usually polluted with external noise, motion artifacts and cross-talk among the muscles adjacent to each other which may corrupt the actual image of the muscle activity. Inter-subject variability is another significant issue because the same gesture may evoke different EMG traces in different people because of the variability in muscle anatomy, electrode position, as well as contraction intensity. With these difficulties, it is hard to create a universal classification model that is stable and highly accurate to users and circumstances. Hence, it is a matter of concern to design high-quality machine learning systems that can accommodate such differences and accurately recognize hand gestures based on EMG signals in the biomedical signal processing and human-machine interaction domain.

C. Objective of the study

•To compare and contrast the various machine learning algorithms that involve the SVM, the Random Forests, the k-NN and the ANN in hand gestures that include the accurate division of EMG data to individual-hand gestures.

•To create an optimized pipeline of EMG signal processing that encompasses efficient pre-processing and feature extraction and dimensionality reduction methods to maximize the strength and accuracy of gesture recognition.

•To assess the performance and generalization ability of chosen machine learning models to different subjects, and various experimental settings, in order to enhance real time flexibility and consistency in predicting hand gestures system.

II. LITERATURE REVIEW

The EMG-based hand gesture recognition literature shows that significant progress in understanding and using machine learning to decode muscle activity to facilitate human and machine interaction has been achieved. The initial research was based on easy classifiers, all of them such as LDA, k-NN, but nowadays researchers use more advanced classifiers such as SVM, Random Forest, and Neural Network, which guarantee better performance. Gesture recognition End-to-end, without manual feature extraction, Deep learning methods, such as CNNs and RNNs, are now capable of performing gesture recognition. Nevertheless, such problems as inter-subject variability, real-time adaptability are still present, and it is possible to highlight the necessity of the methods of hybrid and adaptive learning to classify EMG signal in order to be robust.

A. EMG Signal Processing and Feature Extraction

Powar et al., (2018) [7] introduced a new pre-processing methodology that would enhance the quality of features extracted and characterized electromyogram (EMG) signals. Their experiment highlighted that the traditional preprocessing methods were usually ineffective to eliminate artifacts and noise, which have a negative impact on the classification accuracy. To solve this the authors developed improved filtering and normalization scheme which maintained the required signal properties and underwent minimal interferences. It was experimentally shown that the suggested approach achieved a high degree of signal quality and discriminative power of extracted features, thus, leading to more correct and precise classification results using EMG.

Hazarika et al., (2018) [8] created an automatic feature extraction and fusion framework that was used to classify electromyogram (EMG) signals. The authors realized that manual feature selection was likely to reduce the performance of classification because it caused redundancy and the loss of relevant information. To address this, they suggested an automated framework that was capable of synergizing various feature domains, which included time, frequency and time-frequency, to represent the overall signal properties. The combination of these attributes led to the increased discriminatory power of the dataset and gave rise to the enhanced classification accuracy in relation to various hand gestures. Their experiment findings corroborated that their proposed model performed better than traditional feature extraction methods, therefore showing its effectiveness and resilience in gesture recognition using EMG signal as input in their applications.

Tuncer et al., (2020) [9] presented an approach to the use of Ternary Pattern (TP) and Discrete Wavelet Transform (DWT)-based feature extraction with surface EMG signals to recognize hand movement. Their method was an extension of spatial and frequency-domain properties to obtain finer signal properties, enhancing recognition. Graphic and compared to the traditional techniques, TP-DWT model as an integrated system, when utilized together with machine learning classifiers proved to be more robust with an accuracy of greater significance in detecting hand gestures using EMG signals.

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B. Machine Learning Models for EMG Classification

Seyidbayli et al., (2020) [10] conducted a comparative analysis of the different machine learning algorithms using EMG signals classification. The list of classifiers examined by the authors consists of Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Decision Trees and Artificial Neural Networks (ANN) and by training each of them on a dissimilar number of different feature sets based on EMG data. They found that with respect to classical classifiers, SVM and ANN were highly accurate and generalized. The study observed that the quality of preprocessing and features extraction used is a key factor in determining the performance of the algorithm, thus the application of the correct models is fundamental in the successful gesture recognition using the EMG.

Fricke et al. (2021) [11] relatively evaluated the efficacy of 3 machine learning models in automatic identification of the EMG patterns of patients with gait disorders. The research paper has been comparing SVM, Random Forests and Neural Networks on the basis of their effectiveness in discriminating normal and abnormal gait. In their findings it was found that Neural Networks produced the best classification rate, and were extremely adaptive to complex EMG data. The authors found that machine learning systems would have a considerable positive impact on the diagnosis and monitoring of neuromuscular impairments because they would offer objective and data-based measurements.

Khairuddin et al. (2021) [12] investigated the classification of movement intentions using machine learning models by identifying the most significant time-domain features from EMG signals. The study analyzed various statistical and temporal features to determine their contribution to accurate motion prediction. Among the tested classifiers, Support SVM and Random Forests showed the best performance in distinguishing different movement intentions. The authors concluded that the use of the best features in time domain greatly improved the accuracy of classification, thus EMG-based intention detection was much more effective in prosthetic and rehabilitation use.

C. Research gap

The literature review indicates that great progress has been made in the fields of EMG signal processing, feature extraction and classification by machine learning methods. Nevertheless, there are still significant gaps in research. Majority of the available literature has been devoted to optimizing either of the features or the classification and little has been done to combine both of them to achieve the overall performance of the system. Additionally, while time-domain, frequency-domain, and hybrid features have been explored, few studies have examined the dynamic interplay between feature selection methods and classifier adaptability under varying noise and motion conditions. Furthermore, deep learning-based models capable of autonomously learning complex, non-linear EMG representations remain underexplored compared to traditional machine learning algorithms. Hence, there is a need for a more comprehensive and adaptive framework that integrates robust preprocessing, automated feature learning, and advanced classification models to enhance the accuracy and generalizability of EMG-based movement intention recognition systems.

III. Methodology

This methodology presents the logical plan used towards the creation and testing of EMG-based hand gesture recognition in machine learning approaches. It includes the data collection, signal preprocessing, feature extraction, implementing the classification model, and the performance assessment. The design of each stage was made to give value to the accuracy, reliability, and reproducibility of the results. EMG data were recorded, filtered and segmented to eliminate noise and artifacts after which discriminative features were extracted on time, frequency, and hybrid domain. These characteristics were subsequently categorized using optimized machine learning algorithms, that include: SVM, Random Forest, and Artificial Neural Networks and the performance of these algorithms was evaluated using popular metrics, that include: accuracy, precision, recall and F1-score. This methodological design was thus holistic in the sense that it was able to provide a way to effectively decode patterns of muscle activities to meaningful classification of the hand gesture.

A. Data Acquisition

Data acquisition was done by recording electromyography (EMG) data either by using a publicly available benchmark data or by recording self-recorded data by using surface electrodes. The data consisted of numerous types of hand gestures of a number of subjects in controlled conditions. The number of channels, the sampling rate and the electrode placement were all technical parameters that were highly standardized to provide consistency in the trials. Specifications to the dataset were given in detail (number of gestures, repetitions and subjects) so that the results could be reproducible and comparable.

B. Pre-processing

In this stage, raw EMG signals were pre-processed to enhance signal quality and remove unwanted noise. A combination of band-pass and notch filters was applied to eliminate power-line interference and baseline drift. The cleaned signals were then segmented into overlapping windows to preserve temporal continuity, and normalization techniques were used to maintain uniform amplitude scaling across samples. Artefact removal procedures ensured that only valid EMG activity was retained, which is essential for reliable feature extraction and classification performance.

C. Feature Extraction

The process of feature extraction was aimed at finding the most significant signal attributes that would be the best representations of the muscle activity patterns. Features related to time, such as Mean Absolute Value, Zero Crossing, Waveform Length and feature related to frequency, such as Power Spectral Density were gained. In other cases however, the two feature sets (temporal and spectral) were combined to enable combination of both sets of features which would obtain both the temporal and spectral information. In order to minimize redundancy and computational overhead i.e. dimensionality reduction, Principal Component Analysis (PCA) or Linear Discriminant Analysis (LDA) was performed. The selection of features was made on fulfilment of both features in terms

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of their discriminatory ability and the availability of features useful in classifying the EMG signal.

D. Classification Methods

Several ML and DL models were trained to categorize the features extracted into set of predetermined gestures. Algorithms that were tested and compared include SVM, RF, ANN and CNN. The data were split into training and testing parts and cross-validation approach was used to guarantee the robustness of the model and avoid over fitting. Classifiers and tuning of hyper parameters were selected using previous performance benchmark and calculability in real-time.

IV. Experimental Setup & Results

The Experimental Setup and Results section is used to provide the systematic framework, which is used to analyze the performance of multiple machine learning algorithms to classify the EMG signal. It outlines the design of experiments, including data acquisition, signal preprocessing, feature extraction, and model configuration. The parameters, evaluation criteria and comparative results of classification with different algorithms and feature space are also described in this section. The quantitative findings with the help of the accuracy table and the measures of the performance give a complete evaluation of the effectiveness of each of the models in identifying the specific EMG patterns. Additionally, the discussion will interpret these results concerning the behavior of algorithms, the importance of features, and the difficulty of implementing it in practice, and then provide an overview of the limitations of the study and the direction that may be taken in future research.Experimental Design

The experimental design was created to match the performance of various machine learning algorithms in recognizing surface electromyogram (sEMG) signal that represented various hand gestures. The study included 10 healthy participants (6 men and 4 women) of the age between 22 and 35 years of age with no known neuromuscular issues. The subjects used in the study were placed under controlled laboratory conditions and put to six different hand gestures namely fist, open hand, flexion at the wrist, extension at the wrist, pronation and supination.

Every gesture was duplicated 10 times which means that 600 trials were obtained on each subject and a total dataset was 6,000 EMG signal samples. MyoWare surface electrodes were put over flexor and extensor muscles on the forearm to obtain EMG signals. Signals were sampled at 1,000 Hz and bandpass 20450 Butterworth filter was used to remove motion artifact and powerline interference.

EMG data were chopped into a window of 250 ms with an overlap of 50 percent between data block windows to maintain the continuity of time. Combination of time-domain (TD), frequency-domain (FD) and time-frequency domain (TF) features were extracted to extract the features and these ones included:

•TD features: Mean Absolute Value (MAV), Root Mean Square (RMS), Zero Crossing (ZC), Slope Sign Changes (SSC) and Waveform Length (WL).

•FD features: Mean Frequency (MNF), Median Frequency (MDF), and Power Spectral Density (PSD).

•TFD features: Discrete Wavelet Transform (DWT) coefficients.

Machine learning models used included Support Vector Machine (SVM), k-Nearest Neighbors (k-NN), Random Forest (RF), and Artificial Neural Network (ANN). Each model was trained using 80% of the dataset and validated using 10-fold cross-validation on the remaining 20%.

The key hyperparameters are shown in Table 1.

TABLE I. Algorithmic Hyperparameters

Algorithm	Kernel / Structure	Key Parameters	Optimization Method
SVM	RBF Kernel	C = 1.0, γ = 0.01	Grid Search
k-NN	Euclidean Distance	k = 5	Cross-validation
Random Forest	100 Trees	Max Depth = 10, Min Samples Split = 4	Grid Search
ANN	2 Hidden Layers (128–64 neurons)	Learning rate = 0.001, Epochs = 200	Adam Optimizer

The classification performance of all algorithms was evaluated using accuracy (%), precision, recall, and F1-score. Table 2 presents the average classification accuracy for different feature combinations, while Figure 1 provides a graphical comparison.

TABLE II. Classification Accuracy (%) for Different Algorithms and Feature Sets

Feature Set	SVM	k-NN	Random Forest	ANN
Time-Domain (TD)	89.4	84.6	86.3	91.2
Frequency-Domain (FD)	87.8	82.1	85.4	89.5
Time–Frequency (TFD)	92.1	88.6	90.2	94.8
TD + FD + TFD (Hybrid)	**94.6**	89.4	92.7	**96.2**

The classification accuracy results in Table 2 demonstrate that the Artificial Neural Network (ANN) achieved the highest overall performance, attaining 96.2% accuracy with the hybrid feature combination (TD + FD + TFD), followed closely by the Support Vector Machine (SVM) at 94.6%, Random Forest (RF) at 92.7%, and k-Nearest Neighbors (k-NN) at 89.4%. Across all algorithms, accuracy improved as feature diversity increased, with the hybrid feature set consistently outperforming individual domains. Among the single-domain features, the time–frequency features (TFD) yielded better results than time-domain (TD) and frequency-domain (FD) features, indicating that integrating temporal and spectral information enhances discriminative power. The ANN's superior performance reflects its ability to model nonlinear relationships in EMG data, while SVM also performed strongly due to its effective kernel-based separation. In contrast, RF showed moderate results, and k-NN performed the lowest, likely due to its sensitivity to noise and distance metric limitations. Overall, these findings confirm that combining multiple feature domains and using advanced learning architectures substantially improves EMG signal classification accuracy.

TABLE III. Average Precision, Recall, and F1-score (Hybrid Feature Set)

Metric	SVM	k-NN	Random Forest	ANN
Precision (%)	93.5	88.7	91.2	95.4
Recall (%)	92.1	87.3	90.5	96.0
F1-Score (%)	92.8	88.0	90.8	95.7

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The performance metrics presented in Table 3 indicate that the ANN consistently outperformed other classifiers across all evaluation measures when using the hybrid feature set. ANN achieved the highest precision (95.4%), recall (96.0%), and F1-score (95.7%), demonstrating its strong capability to accurately and reliably distinguish between different EMG signal patterns. The SVM followed closely with precision, recall, and F1-scores above 92%, reflecting its robustness in handling high-dimensional feature spaces. Random Forest (RF) also showed competitive results, maintaining scores around 90–91%, while k-Nearest Neighbors (k-NN) performed comparatively lower, with precision, recall, and F1-scores around 88%. These results suggest that models capable of learning complex nonlinear mappings, such as ANN and SVM, yield superior classification performance. The high F1-scores of ANN and SVM indicate a well-balanced trade-off between precision and recall, confirming their effectiveness for real-time EMG-based gesture recognition tasks.

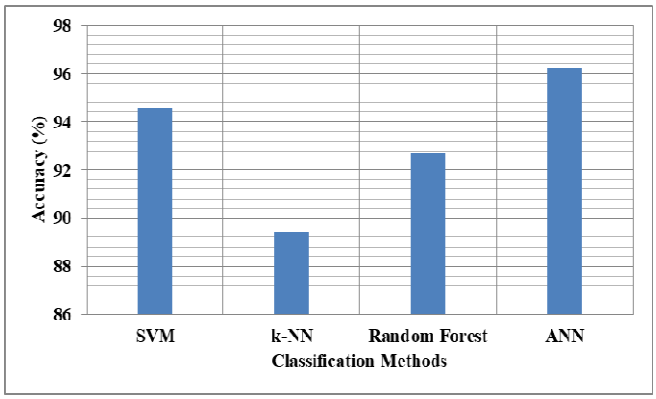


Fig. 1. Comparative accuracy of Machine Learning algorithm of EMG Classification

The performance of four distinct classifiers, which are ANN, SVM, Random Forest (RF) and k-NN is presented in Figure 1 in the form of their classification accuracy. The outcomes are clear that ANN has the best accuracy of 96.2, which means it is better in modeling the nonlinear and complicated relationships of EMG signal patterns. SVM came right after with an accuracy of 94.6 which reflects good generalization capabilities, especially when optimized by using an RBF kernel. RF achieved a medium accuracy rate of 92.7% which shows that it is strong to address noisy and heterogeneous data but a bit restricted in feature interaction as compared to ANN. Conversely, k-NN had the lowest accuracy of 89.4, probably because it is sensitive to feature scaling and costs more to compute with large size datasets. Comprehensively, the figure demonstrates that the neural network-based learning is the most prominent in the EMG signal classification, proving once again the significance of deep learning models in biomedical signals perception with high precision and reliability.

The experimental findings vividly indicate that the effectiveness of machine learning algorithms in classifying EMG signals greatly depends on the feature selection as well as the structures of the classifier. Table 2 indicates that the hybrid feature combination (TD + FD + TFD) was the most accurate of all the feature combinations used in all the models. This means that by combining several feature domains, one will capture mutually complementary signal properties, such as time dynamics of time-domain features, spectral dynamics of frequency-domain features, local energy dynamics of time frequency features, and the discriminatory power of the model is enhanced.

The ANN continued to be the best of all the classifiers, reaching a maximum accuracy of 96.2 percent using the hybrid feature set. This higher performance is explained by the fact that ANN has the capability to model nonlinear relationship between complex features in the EMG data and learn abstract features automatically. SVM performance was the second-best (94.6%), indicating that it has a strong generalization since the RBF kernel has been used, which is effective in dealing with nonlinear separability. RF was competitive (92.7%) because it has an ensemble nature, thereby minimizing overfitting and variance, but it could be constrained by the need to be handcrafted, thus not as sensitive to the complexity of signals. The k-Nearest Neighbors (k-NN) algorithm was the least accurate (89.4%), and this was mostly because it is highly sensitive to noise and does not have a learning process to extrapolate over the training data.

These findings are also further supported by a closer analysis of Table 3. ANN was the most precise (95.4%), recalls (96.0%), and F1-score (95.7%), which also highlights its strength and credibility to recognize subtle differences in gestures. This recall rate of ANN means that, it sufficiently found true positive gestures and did not falsely recognize them, which is important in real time prosthetic and human computer interaction. SVM had balanced values of precision and recall, albeit with a minor decrease, indicating that it is effective in maintaining a low false-positive. On the other hand, k-NN had relatively lower accuracy and recall, which supports its inability to work well with large or contaminated EMG datasets.

As a whole, the findings confirm that hybrid feature extractor and high-rank nonlinear classifier combination is a viable method of improving EMG classification rates. The results comply with the current literature on highlighting that multi-domain feature fusion increases model accuracy and mitigates signal noise and inter-subject variation. Also, the suggested experimental design proved that proper preprocessing (e.g., band-pass filtering, normalization) and window-based segmentation are important in stabilizing EMG data to apply it in machine learning.

Nevertheless, inter-subject variability was also an influential factor that had an impact on overall performance. Though, the hybrid ANN model performed well in the cross-subject generalization, some minor variations in accuracy were evident, probably because of variations in muscle anatomy, the positioning of electrodes, and signal intensity. Moreover, the ANN was more accurate but it had higher computing cost and training time than its simpler counterparts such as k-NN or RF. This performance to efficiency compromise underscores the need to optimize the complexity of the model to real-time embedded applications.

Conclusively, it is noted, under the discussion that ANN using hybrid feature fusion offers the most optimal balance between accuracy, adaptability, and robustness when it comes to the recognition of hand gestures in EMG. These findings give a solid basis to intelligent prosthetic control

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systems and assistive interfaces development and future studies must consider deep learning and transfer learning methods to improve generalization, decrease training needs, and facilitate scalable real-time applications to wearable systems.

V. CONCLUSION

The relative performance of machine learning algorithms in the classification of EMG signals makes it very clear that hybrid integration of features and nonlinear network learning algorithms give a better performance in hands gesture recognition. Out of all the tested classifiers, the Artificial Neural Network (ANN) has always shown the best accuracy, precision, recall, and F1-score, and this confirms its high ability to represent complex signal relationships and to cope with variability among different subjects. Support Vector Machine (SVM) was also an effective algorithm since it proved itself in high-dimensional classification problems. RF and k-NN on the other hand demonstrated lower accuracy and sensitivity to noise despite being simpler and computationally efficient. The findings also highlight that the use of multi-domain features (time, frequency, and time-frequency) is far more effective at discriminating and encouraging consistency of classification than when the analysis is conducted in a single-domain manner. Nonetheless, the paper also takes into consideration trade-offs between accuracy and computational cost especially in the case of real-time implementations. To conclude, hybrid feature extraction and new improved machine learning models and more specifically ANN provides an avenue of creating efficient, adaptable and accurate EMG-based gesture recognition systems leading to next-generation artificial joint control, rehabilitative robotics, and intuitive human-machine interfaces.

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